Does paracetamol given after the measles-rubella vaccine lower the antibody response in nine-month-old infants in Jaipur

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Introduction

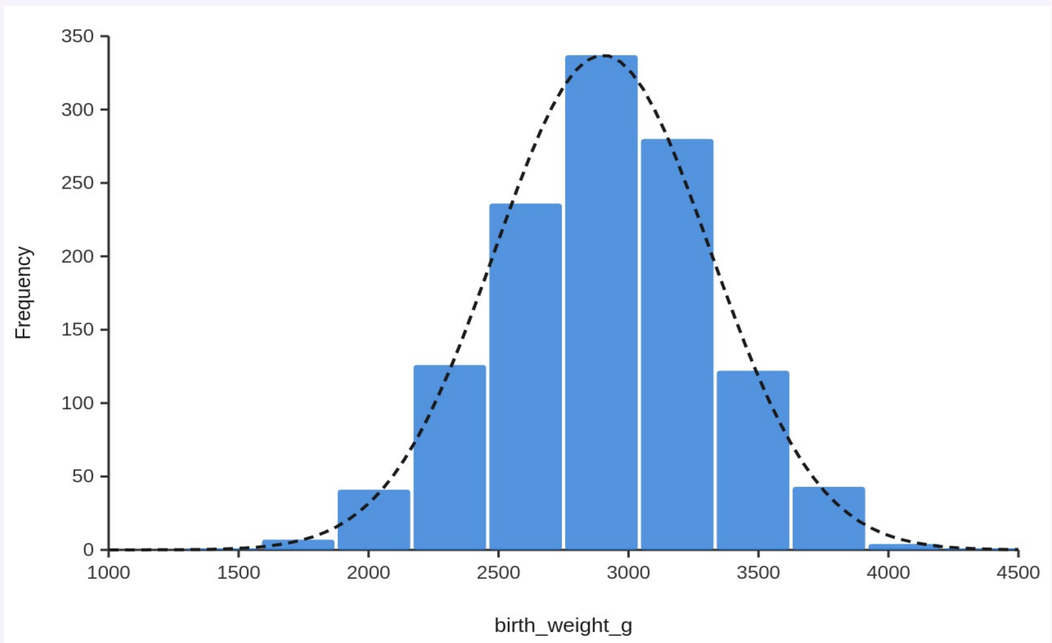
- Measles still kills children where vaccine coverage is low, and two doses of a measles-containing vaccine prevent almost all cases (Moss & Griffin, 2012).
- In a randomised trial, infants given prophylactic paracetamol at vaccination had lower antibody responses to several vaccines (Prymula et al., 2009).
- Trials in Europe found lower antibody responses when infants were given paracetamol at vaccination, but a search of OpenAlex and PubMed for 2009 to 2025 found no Indian study of measles antibody…
- The present study asked: Does paracetamol given after the measles-rubella vaccine lower the antibody response in nine-month-old infants in Jaipur?
- H1: Infants given paracetamol after MR1 will seroconvert less often than infants not given paracetamol.
- H2: Seroconversion will not differ between boys and girls.

Methods

- Design. This was a cohort study.
- Participants. Infants born in the simulated Janani cohort, Jaipur, who received the first measles-rubella dose (MR1) at 9 months.
- Measures. Seroconversion. Recorded as yes or no.
- Paracetamol after MR1. Recorded as yes or no.
- Infant sex. Recorded as: Boy, Girl.
- Birth weight. Taken from the hospital laboratory report or the antenatal card.
- Low birth weight. Recorded as: Yes, No.
- Procedure and ethics. The data are simulated to test Appuscholar; no real infant took part.
- Analysis. The main hypothesis was tested with a logistic regression.

Results

- Results for each hypothesis follow.
- A chi-square test of independence showed a significant association between paracetamol after MR1 and seroconversion, $\chi^2(1, N = 1133) = 11.42$, $p < .001$, Cramér's V = 0.10.
- A chi-square test of independence showed no significant association between infant sex and seroconversion, $\chi^2(1, N = 1133) = 0.05$, $p = .827$, Cramér's V = 0.01.
- A logistic regression with birth weight (g), infant sex: Girl (vs Boy), paracetamol after MR1: Yes (vs No) as predictors of seroconversion (Yes versus No) was significant, $\chi^2(3, N = 1133) = 10.72$, $p = .013$, Nagelkerke $R^2 = .021$. The model classified 91.4% of cases correctly (AUC = .59). Odds ratios: birth weight (g) (OR = 1.00, 95% CI [1.00, 1.00], p = .474); infant sex: Girl (vs Boy) (OR = 0.97, 95% CI [0.64, 1.48], p = .902); paracetamol after MR1: Yes (vs No) (OR = 0.46, 95% CI [0.29, 0.73], p < .001).
- There was a moderate positive correlation between the mother's haemoglobin at 34 weeks and birth weight (g), which was statistically significant, $r(1196) = .31$, $p < .001$, 95% CI [.26, .36].
- An independent-samples Welch t-test showed that birth weight (g) differed significantly between boys (M = 2955.67, SD = 404.24) and girls (M = 2840.56, SD = 415.40), $t(1183.18) = 4.85$, $p < .001$, g = 0.28, 95% CI of the difference [68.59, 161.64].
- Distribution of birth weight (g)
- Note. The dashed curve is a normal distribution with the same mean and standard deviation as the data.



Discussion

- Infants whose parents gave paracetamol after MR1 seroconverted less often, which supports H1 and follows the same direction as the trial of prophylactic paracetamol (Prymula et al., 2009).
- Paracetamol was given more often to infants with fever, so fever itself or a stronger reaction may explain part of the link. The data are simulated, and paracetamol use was reported by parents.

Conclusion

- Paracetamol given after MR1 went with lower seroconversion. Advice at the vaccination session should say to give it for fever, not to every infant.

References

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